Summary in Graph

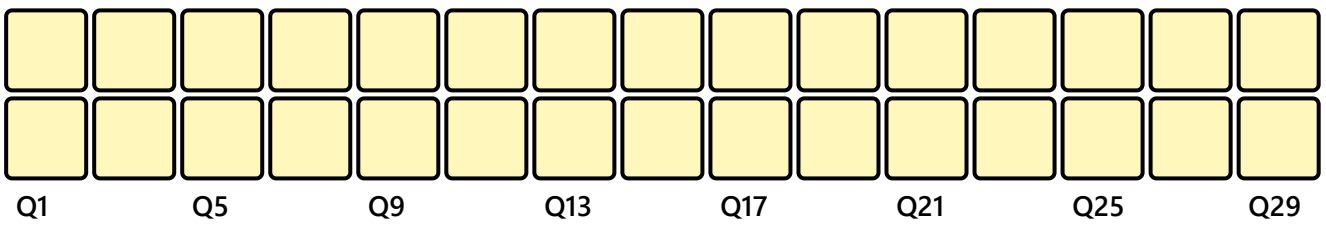
Exam Summary (GATE Overflow CS Test Series 2026 | Discrete Mathematics | Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (5) out of 45 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Combinatory

The minimum number of people that must be in a room to ensure that at least three were born on the same day of the week is _____

Your Answer: | Correct Answer: 15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

Consider a set A with 6 elements. Let N_1 denote the number of bijective functions from A to A and let N_2 denote the number of onto (surjective) functions from A to A. $N_2 - N_1 =$ _____.

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

Number of bit strings of length 10 that do not end in "111" is ____

Your Answer: | Correct Answer: 896 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

Let $G(x) = \frac{1}{(1-x)} = \sum_0^\infty f(i)x^i$ where $|x| < 1$. $f(i)$ is

A. 1
B. i
C. $2i$
D. $i+1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

The number of bit strings of length 6 that do not contain "1111" as a substring is ____

Your Answer: | Correct Answer: 56 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Numerical Type | Award: 1 | Penalty: 0 | Combinatory

The number of positive integers less than 1000 which are divisible by 7 but not by 11 is ____

Your Answer: | Correct Answer: 130 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

How many 10 digit numbers greater than 1987654321 have no two digits same?

A. $8 \times 9!$
B. $9 \times 9!$
C. $7 \times 9!$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type Award: 1 Penalty: 0.33 Combinatory

How many bit strings of length 7 contain more 0's than 1's?

A. 93
B. 84
C. 64
D. none of these

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Combinatory

15 students took a quiz of 15 questions where each question carry 1 mark with no negative markings for wrong answer. The sum of their scores came out to be 100. Now consider the following two statements

- S_1 : All the scores are different.
- S_2 : There are at least two students having the same score.

Which of the above statements is/are TRUE?

A. Only Statement S_1 is true
B. Only Statement S_2 is true
C. Both S_1 and S_2 can be true
D. Both S_1 and S_2 are false

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Choice Type Award: 1 Penalty: 0.33 Combinatory

Let $S_1 = \{1, 2, 3, \ldots, 100\}$ and $S_2 = \{a, b, c, \ldots, z\}$. Number of one-one (injective) functions from $S_2 \rightarrow S_1$ when $f(m) = 13$ is ___

A. ${}^{100}P_{25}$
B. ${}^{99}C_{25}$
C. ${}^{99}P_{25}$
D. 1

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #11 Numerical Type Award: 2 Penalty: 0 Combinatory

Professor Nalanda has 6 distinct properties which he decided to distribute among his 4 daughters. He hires a consultancy firm for the same which charges Rs. 100 for each of the possible distribution assuming each daughter gets at least one property and all daughters being considered distinct. How much fee will Professor have to pay the agency (in multiples of 100)?

Your Answer: Correct Answer: 1560 Not Attempted Time taken: 00min 00sec Discuss

Q #12 Numerical Type Award: 2 Penalty: 0 Combinatory

A game of cards is played among 4 friends. After the initial split of 52 cards, the tallest person told that he has got 4 Aces and all the remaining 9 cards are of the same suit. Total number of ways in which this is possible is ____

Your Answer: | Correct Answer: 880 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

For T20 Cricket Word Cup $2021, 30$ probables are selected with 5 from West, 7 from North, 8 from South, 6 from East and 4 from Central India with each zone having a captain. From these the final squad of 15 will be selected. What is the probability that all the zones will be represented in the final squad by their captains?

A. $\frac{5\times {}^{25}C_{10}}{{}^{30}C_{15}}$
B. $\frac{5\times {}^{25}P_{5}}{{}^{30}C_{10}}$
C. $\frac{5\times 7\times 8\times 6\times 4}{{}^{30}C_{15}}$
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

Match the sequences on the left with their corresponding generating functions on the right

$A.$	$0, 1, 2, 3, 4, 5, 6, 7, \ldots$	$P.$	$\frac{x(1+x)}{(1-x)^3}$
$B.$	$1, 1, 1, 1, 1, 1, \ldots$	$Q.$	$\frac{x}{(1-x)^2}$
$C.$	$1, 1, \frac{1}{2!}, \frac{1}{3!}, \frac{1}{4!}, \ldots$	$R.$	$\frac{1}{1-x}$
$D.$	$0, 1, 4, 9, 16, 25, \ldots$	$S.$	e^x

A. $A-P, B-Q, C-R, D-S$
B. $A-P, B-R, C-S, D-Q$
C. $A-Q, B-R, C-S, D-P$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Number of positive integer solutions to the equation $4x + y + z = 21$ is _____

Your Answer: | Correct Answer: 40 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

A board is formed by combining 100 squares of same size, 10 in each direction. The total number of squares (of all sizes) that can be formed from the resultant board is _____

Your Answer: | Correct Answer: 385 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

The generating function of the sequence $1, 1, 2, 3, 5, 8, \ldots$ is $\frac{1}{1-x-x^2}$. The sequence generated by generating function $\frac{1}{1-3x-9x^2}$ is _____

A. $1, 3, 18, 81, 405, \ldots$
B. $1, 0, 3, 0, 18, 0, 81, 0, 405 \ldots$
C. $3, 3, 6, 9, 15, 24, \ldots$
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

The number of positive integers less than 1000 which are divisible by exactly one of 7 or 11 is _____

Your Answer: | Correct Answer: 208 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Let $S_1 = \{1, 2, 3, 4, 5\}$ and $S_2 = \{a, b, c\}$. Number of into functions from $S_1 \rightarrow S_2$ is ____

Your Answer: | Correct Answer: 93 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Number of positive even integers less than 1000 having distinct digits is _____

Your Answer: | Correct Answer: 373 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

The number of non-empty partitions of a set of size 5 is _____

Your Answer: | Correct Answer: 52 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Combinatory

The number of ways in which we can choose 3 numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ so that more odd numbers are chosen than even is _____

Your Answer: | Correct Answer: 50 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

Which of the following is the generating function corresponding to the sequence $0, 1, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0, 5, \ldots$.

A. $\frac{x^3}{(1-x)}$
B. $\frac{x}{(1-x)^2}$
C. $\frac{x}{(1-x^3)^2}$
D. $\frac{x}{1-x^3}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

The generating function corresponding to the sequence with closed formula for its nth term as $a_n = 4(6^n) + 7(-2)^n$ is _____

A. $\frac{4}{1-6x} + \frac{7}{1+2x}$
B. $\frac{4}{1+6x} + \frac{7}{1-2x}$
C. $\frac{6}{1-4x} + \frac{2}{1+7x}$
D. $\frac{6}{1-4x} + \frac{2}{1-7x}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

The closed form solution for the nth term $(n > 1)$ of the sequence with generating function $\frac{9x}{1-2x} + \frac{x}{1-x}$ is given by

A. $1.2^n + 1$
B. $2.9^{n-1} + 1$
C. $9.2^n + 1$
D. none of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

In a college 10 Servers are bought and they are to be distributed among 4 laboratories such that each laboratory gets at least one server. The total number of ways in which this can be done is ____

Your Answer: | Correct Answer: 84 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Combinatory

What is the middle term in the expansion of

$$\left(\frac{y\sqrt{x}}{2} - \frac{2}{x\sqrt{y}}\right)^{12}$$

.

A. $924\left(\frac{y}{x}\right)^6$

B. $924\left(\frac{y}{x}\right)^3$
C. $2^{10}\left(\frac{y}{x}\right)^6$
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

The nth term of the sequence generated by the function $\frac{2}{(1-x)^2} \cdot \frac{x}{1-x-x^2}$ is represented as a_n. The value of a_6 is_____.

Your Answer: | Correct Answer: 92 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

A decimal number is called "increasing" if each digit is greater than the previous one (e.g., 3689 is one). The total number of 6 digit increasing numbers is _____

Your Answer: | Correct Answer: 84 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Combinatory

Number of triangles that can be formed with vertices on a 3×3 grid of points is _____

Your Answer: | Correct Answer: 76 | Not Attempted | Time taken: 00min 00sec | Discuss

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